

The Transformation of Writing with Generative Artificial Intelligence: The Effect of Automated Feedback on the Writing Process of EFL Students*

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Abstract

This study examines how a generative artificial intelligence (GAI)-supported Automated Writing Evaluation (AWE) system influences EFL students' writing experiences. The ChatGPT-based system, integrated into Moodle, provided students with scoring and feedback on their writing tasks. Using a qualitative case study design, data were collected through semi-structured interviews with students and a writing instructor, and were analyzed thematically. Findings indicated that the system supported improvements in students' technical writing skills, including grammar, punctuation, and sentence structure, depending on the level of engagement they demonstrated. However, the feedback was primarily surface-level and lacked depth, while teacher feedback was more detailed, context-aware, and easier to understand. Some contextual and pedagogical mismatches limited the system's effectiveness in the EFL context. Despite these challenges, the system served as a helpful complementary tool for drafting, self-practice, and independent learning. The study emphasizes the importance of incorporating teacher input into GAI-supported systems to ensure pedagogical relevance and meet learner needs.

Keywords: EFL learners, feedback perception, qualitative research, automated writing evaluation, generative artificial intelligence

Üretken Yapay Zekâ ile Yazmanın Dönüşümü: EFL Öğrencilerinin Yazma Sürecinde Otomatik Geri Bildirimin Etkisi

Öz

Bu çalışma, üretken yapay zekâ (ÜYZ) destekli bir otomatik metin değerlendirme (OMD) sisteminin EFL öğrencilerinin yazma deneyimlerine etkisini incelemektedir. Moodle'a entegre edilen ChatGPT tabanlı sistem, öğrencilere yazma görevlerinde puanlama ve geri bildirim

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sağlamıştır. Nitel durum çalışması kapsamında öğrenciler ve bir yazma eğitmeniyle yarı yapılandırılmış görüşmeler yapılmış, veriler tematik analizle değerlendirilmiştir. Bulgular, sistemin öğrencilerin gramer, noktalama ve cümle yapısı gibi teknik yazma becerilerini kullanım düzeylerine bağlı olarak geliştirdiğini göstermiştir. Ancak geri bildirimlerin çoğu yüzeysel olup yeterli açıklama içermemiştir. Öğretmen geri bildirimlerinin daha ayrıntılı, bağlama uygun ve anlaşılır olduğu görülmüştür. Bazı bağlamsal ve pedagojik uyumsuzluklar sistemin EFL bağlamındaki etkililiğini sınırlamıştır. Buna rağmen sistem, taslak oluşturma, bireysel çalışma ve ders dışı öğrenme için tamamlayıcı bir araç olarak yararlı bulunmuştur. Sonuç olarak, ÜYZ tabanlı sistemlerin pedagojik uygunluğu artırmak için öğretmen katkısıyla tasarlanması önerilmektedir.

Anahtar Kelimeler: EFL öğrencileri, geri bildirim algısı, nitel araştırma, otomatik yazma değerlendirme, üretken yapay zekâ

Introduction

Writing skills are one of the four core skills in language teaching and play a critical role in students' academic and professional development (Kellogg & Raulerson, 2007). In particular, English writing skills have become a fundamental tool for information sharing and international communication in the digital world. This skill enables students to structure their thoughts, produce knowledge, and communicate effectively, giving them a significant advantage in their international careers (Hyland, 2003; Liu et al., 2021). According to Hyland (2003), individuals need strong writing skills to effectively communicate their knowledge and ideas through global digital networks. In this context, developing students' English writing skills enhances their language proficiency and enables them to actively participate in international academic and professional environments (Fu et al., 2019; Lan, 2015; Hyland, 2003). Despite all these advantages, teaching writing skills is considered one of the most complex and challenging areas of language education (Uyar & Büyükahıska, 2025; Zhao, 2022).

Teachers, especially in crowded classrooms, struggle to effectively teach writing skills. Since writing requires individual feedback and continuous guidance, large classes, time constraints, and resource limitations limit the support that can be provided to students (Hyland, 2003; Harmer, 2004; Toscu, 2025). The assessment of written texts also poses a separate challenge. This process is time-consuming and labour-intensive (Uyar & Büyükahıska, 2025). Furthermore, providing consistent and objective feedback is not always possible (Lee, 2023; Ferris, 2003). Grades given by human evaluators often contain subjectivity, which may not contribute sufficiently to student development (Zupanc & Bosnić, 2020). As a result, this situation can lead to burnout among teachers and negatively affect the quality of feedback (Uyar & Büyükahıska, 2025).

To address these issues in the teaching of writing skills, research on automated writing assessment systems has been conducted since the 1960s (Dikli, 2006). In the literature, various terms are used to describe the automated assessment of writing.

Automated Essay Scoring (AES) is generally defined as the automatic scoring of written texts using natural language processing and machine learning techniques based on predefined criteria and/or statistical models (Attali & Burstein, 2006; Dikli, 2006). While AES typically refers only to scoring, Automated Writing Evaluation (AWE) is used as a broader term that includes not only scoring but also automated feedback, such as comments, suggestions, and error highlights, within a classroom or learning context (Warschauer & Grimes, 2006). Automated Essay Assessment/Evaluation (AEE) and Automated Writing Scoring (AWS) are also sometimes used synonymously with AES, or, in some studies, as umbrella terms (Shermis & Burstein, 2013). Automated Essay Scoring (AES), as a subcategory of AWE, refers to a technological system developed through the integration of disciplines such as natural language processing, computer science, and latent semantic analysis, which analyzes written texts through computer programs to perform evaluation and grading functions (Huawei & Aryadoust, 2023; Shermis & Burstein, 2013). Recent advancements in artificial intelligence technologies, particularly the integration of generative artificial intelligence into education, have significantly accelerated applications in this field (Murphy et al., 2023).

With the release of ChatGPT, a generative artificial intelligence (GAI) tool, in 2022, the concept of GAI has become more widely known. GAI refers to artificial intelligence systems that can generate meaningful, contextually appropriate, and original content using deep learning and artificial neural networks, possessing human-like language capabilities (Bozkurt, 2023). GAI tools benefit writing tasks such as idea generation, language translation, draft development, and consistent text production (Li et al., 2024). GAI-based automated assessment systems are emerging as an important alternative for evaluating writing skills. GAI-supported AWE systems/tools have strong potential to reduce teachers' workload, speed up the assessment process, and provide students with more consistent feedback (Geng & Razali, 2022). These systems can effectively identify grammar errors, suggest corrections, and improve the technical accuracy of written texts (Heintz et al., 2022; Burstein & Leacock, 2006). However, when the feedback provided by AWE systems focuses solely on technical accuracy, it may not adequately support higher-level writing skills such as content consistency, depth of thought, and text organisation (Tambunan et al., 2022). Hyland (2003) and Ferris (2003) emphasise that effective feedback should address linguistic accuracy and students' comprehension skills.

Over the past two decades, global research on AWE systems have examined these systems' validity, reliability, and pedagogical benefits from various perspectives. Early studies reported that rule-based systems such as Criterion® and e-rater® achieved high correlations with human evaluators (Attali & Burstein, 2006; Shermis & Burstein, 2013). With the recent emergence of generative artificial intelligence, the focus has shifted to deep learning and large language models (LLMs). LLM-based AWE systems highlight the superiority of their ability to understand complex language patterns and provide consistent, accurate feedback Heintz et al., 2022;

Burstein & Leacock, 2006; Chan et al., 2020). A study conducted with 18 EFL students in Indonesia revealed that GAI-supported tools such as ChatGPT, Grammarly, and Quillbot are effective in improving students' 'feedback literacy'; however, students still require teacher guidance to understand the underlying reasons behind their errors (Gozali et al., 2024). Research also focuses on hybrid model approaches; studies using teacher and AWE systems feedback have shown that more specific and student-centred feedback is provided (Woodworth, 2022). A different study following a hybrid approach found that systems feedback was effective in terms of grammar but insufficient in terms of content and consistency; it was noted that these shortcomings could be addressed with teacher support (Luo et al., 2025). In summary, global findings indicate that systems provide high accuracy in technical corrections, and LLM-based tools, including ChatGPT, are increasingly prominent in both scoring and feedback generation; however, they still require human teacher input to provide explanatory, contextual, and higher-level writing skill-supporting feedback.

GAI-supported writing tools have also begun attracting attention in Turkey recently. Studies in this context generally examine the effects of these tools on students' writing performance, perceptions, and language skill development (Öğüt, 2022). Selvi (2024) addressed the effects of GAI tools on academic writing at a more general level, emphasising that these technologies offer advantages in terms of technical accuracy and content consistency but that excessive use may harm higher-level skills such as originality and critical thinking. An exploratory study by Ertok (2025) found that ChatGPT positively affected writing, vocabulary, and communication confidence in functional English practice. On the other hand, in a mixed-methods study conducted by Ayan (2025), teacher, GAI, and hybrid feedback types were compared; it was found that hybrid feedback offers a more balanced and practical approach to developing lower- and higher-level writing skills in student writing. The common finding of these studies is that GAI-supported writing tools have strong potential to improve the writing skills of students learning English as a second language when carefully structured and supported by pedagogical guidance.

Within the scope of the research, an existing GAI-based (Virtual Writing Tutor) system was implemented in a real classroom environment using a plugin for the Moodle learning management system. Moodle is known as an open-source LMS (Learning Management System). An LMS is a software platform designed to plan, deliver, manage, and evaluate educational content. It facilitates course creation, student progress tracking, and communication by creating an interface between students and educators (Rutter & Matthews, 2002). LMS is expected to provide students with immediate feedback on written texts and save teachers time in the assessment process. Within the scope of the research, students' interactions with the system, their perceptions of feedback, and their experiences of development in writing skills were examined in depth through qualitative data. This study aims to reveal the theoretical effectiveness of system and user experiences regarding their use in a real-world context.

Research Objective

This study aims to investigate the potential of GAI-supported system in addressing challenges encountered in teaching writing skills. Within the scope of this study, the contributions of system will be evaluated in addressing issues such as the difficulty of providing feedback, subjectivity in evaluation, and time constraints faced by instructors, especially in large classrooms. The impact of GAI-supported tools, which evaluate written texts objectively, more quickly, and more effectively, on the quality of writing education has been investigated.

Research Rationale

Writing skills are one of the fundamental components of language learning and directly affect students' academic achievement and professional development. However, teaching writing skills is a significant burden for teachers in terms of time and resource requirements. The challenges of providing individual feedback in large classrooms can prevent students from receiving adequate development opportunities. In this context, GAI-supported assessment systems can be an important tool to reduce teachers' workload and provide students with more personalised and regular feedback. This study is expected to make writing education processes more efficient, inclusive, and sustainable.

Research questions

1. How does GAI-supported system affect the development of technical writing skills in EFL students; how do the strengths and limitations of these systems differ according to student and teacher experiences?
2. What contributions does GAI-supported system make to the writing process of EFL students, and how do these systems provide the feedback perceived and utilised by students?
3. What contributions does GAI-based system make to the writing instruction when used with teacher feedback?
4. What are the students' and teachers' suggestions for the design of GAI-supported system

Method

Design

This research was conducted using the case study design, one of the qualitative research methods. The study examined the effect of GAI-supported system on the writing skills of students learning English as a foreign language, as well as the opinions, experiences, and evaluations of students and teachers regarding using this

system. The case study examined the integration of the system into the learning environment within the scope of a writing course at a state university.

Automated Writing Evaluation System Design

Within the scope of the research, the Moodle Plugin, the open-source learning management system of the Virtual Writing Tutor software, was used. In this direction, a hosting service was rented to provide access to students, and a special domain name was obtained to create a Moodle-based web platform. Thus, a learning environment was prepared that students could easily access online. The (VWT) Moodle plugin can only perform text-based evaluation and cannot evaluate texts written by students on paper. The requirement for students to create their texts in a digital environment has limited the measurement of writing skills, especially in face-to-face education. Additionally, the system does not support recognizing visual content (e.g., handwriting, photos).

Figure 1. Assignment upload interface view

This study analyzed only writing assignments submitted in a digital environment and revisions automatically saved by the system. If students edited their writings in the system and closed the window without clicking the ‘Save Changes’ button, the changes were not reflected in the database and therefore could not be included in the analysis. Figure 1 shows the interface where students upload their written texts to the system. Through this component, students submit their writing assignments, which

The screenshot displays a web-based writing interface. At the top, it says 'Narrative Paragraph'. Below this, there are dates: 'Açıldı: Tuesday, 19 March 2024, 12:00 AM' and 'Son tarihi: Thursday, 21 March 2024, 9:33 AM'. A 'Tamamla' button is visible. The main area is a large text box labeled 'Çevrimiçi metin' (Online text). Above the text box is a rich text editor toolbar with various icons for text formatting (bold, italic, underline, etc.) and alignment. Below the text box, there are three buttons: 'Automated evaluation', 'Değişiklikleri kaydet' (Save changes), and 'İptal' (Cancel).

are automatically evaluated by the system. Feedback is provided immediately after the assignment is submitted.

Assignment Score: 48.33

Statistics

1. You have written 117 words.
2. I count a total of 1 paragraph.
3. You have written 0 question.

Target structures: 20%

Target structures 1:

You are off topic. You have not used any of these words to tell a story about a dream: *had a dream, I woke up, I think my dream means, I think it means*. Your score is 0%.

made

Target structures 2:

When people talk about events in the past, they usually use common past tense verbs. You haven't used many. Use more for a higher score. Your score is 60%.

Target structures 3:

We learned some past tense negative verbs, so I expected to see some of these words in your story: *didn't, wasn't weren't, couldn't, wouldn't, did not, was not, were not, could not, would not*. You didn't use any. Your score is 0%.

Development: 25%

Length:

Your text is a little short. Write more. Your score is 50%.

Paragraphs:

You have 1 paragraph instead of 2. When you change topic, change paragraphs. Your score is 0%.

Language Accuracy: 100.00

Number of errors: 0

Feedback: Your text is error-free. That's very good.

Figure 2. Sample scoring and feedback screen

Figure 2 shows an example of the feedback screen in the Virtual Writing Tutor system. This system provides students with an overall score and detailed diagnostic feedback on their assignments. The system evaluates the text based on various criteria, such as word count, paragraph structure, use of target grammar structures, development, and language accuracy. In the example shown, the student has received an excellent language accuracy score but significantly lower scores for development and target structure usage. The system highlights specific linguistic features such as the absence of past tense verbs or paragraph division and provides short comments and percentage-based scores. This visual, structured feedback helps students identify weaknesses in their writing and revise accordingly, supporting a more autonomous, data-driven writing process.

Participants

The participants in this study consisted of 53 students enrolled in an English preparatory programme at a state university in the 2023-2024 academic year and a faculty member responsible for writing courses. Nine students were selected for in-depth interviews through purposive sampling, considering their level of interaction with the system and the diversity of their writing experiences. During the six-week implementation period, students were divided into three groups based on their frequency of system use: advanced, intermediate, and novice users. This classification was determined based on how often students uploaded texts to the system, their level of interaction with feedback, system logs, and their statements. Factors such as system usage frequency, academic achievement, and voluntariness were considered to ensure

participant diversity and data richness. This sampling strategy allows for analyzing different user profiles and writing development paths. Table 1 shows the classification of students according to their usage levels, the number of applications they participated in, and the frequency distribution at the group level.

Table 1
Student Participation in the System by Usage Category

Usage Category	Number of Sessions Attended	Student Codes	n	%
High	5–6 sessions	S1, S2, S3	3	33.3%
Medium	3 sessions	S4, S5, S6	3	33.3%
Low	1 session	S7, S8, S9	3	33.3%
Total	—	—	9	100%

Materials and Tools

The study was conducted as part of the Writing II course in the English preparatory programme at a state university. The course consists of a six-week process in which narrative and descriptive paragraph types are taught. The general structure of the course included assigning homework, having students write paragraphs at home, and providing feedback in class. Students were allowed to use the productive AI-supported VWT tool while writing their paragraphs at home or in class.

Data Analysis

Data were collected through a semi-structured interview form developed by the researcher. The interview questions aimed to reveal the experiences of students and faculty members with the system, their perceptions of feedback, the changes they observed in their writing skills, and their opinions on the use of the system. The interviews were conducted one-on-one during the autumn semester of 2024 with the participants' consent and were recorded. Each interview lasted an average of 10-15 minutes and was transcribed verbatim to prepare it for analysis. Research ethics principles were followed in the study, and all participant information was kept confidential.

The collected data were analysed using the six-stage thematic analysis method proposed by Braun and Clarke (2006). Table 2 presents the stages and what was done in each stage.

Table 2

Phases of Thematic Analysis (Adapted from Braun & Clarke, 2006)

Step	Original Term (Braun & Clarke)	Definition	Application in This Study
1	Familiarization with the data	Reading the data closely to identify initial patterns of meaning.	Student and expert interviews were transcribed in Word format and read multiple times by the researcher. Noteworthy statements were highlighted.
2	Generating initial codes	Converting meaningful data segments into codes.	In addition to the predetermined code list, new expressions emerging from the data were labeled using open coding. Coding was guided by the research questions.
3	Searching for themes	Grouping similar codes into potential themes.	Similar codes were clustered and themes were formed. Student and expert data were analyzed separately and later integrated.
4	Reviewing themes	Ensuring that themes align with the coded data and entire data set.	Each theme was reviewed to ensure it was supported by relevant codes. Weak or overlapping themes were merged or removed.
5	Defining and naming themes	Clarifying and naming the core essence of each theme.	The meaning of each theme was clearly defined, theme descriptions were written, and relevant participant quotes were selected.
6	Producing the report	Presenting findings with thematic structure supported by data excerpts.	Themes were systematically presented in relation to the research questions and supported with direct quotations in the findings section.

The coding process was carried out using MAXQDA software. To increase the reliability of the codes, a second researcher performed cross-checking and ensured thematic consistency. Separate theme clusters were created for each research question. One of the researchers was also responsible for the technical aspects of the system used in the study and was its implementer. This situation carries a potential risk of subjectivity in data analysis. To mitigate this limitation, the researcher took reflective notes throughout the analysis process and emphasised the importance of continuously reviewing their position (Creswell & Poth, 2018). The hierarchical theme-code structure developed during the analysis is presented in Figure 3 to visually demonstrate the organization of themes and codes generated during the coding phase.. Table 3 contains explanations of the main themes.



Figure 3. Coding system

Table 3
Thematic Codes and Explanations

Theme	Description
Development of Technical Writing Skills	The supportive or limiting effects of the AI-based system on learners' technical writing skills.
Perceptions of the Nature of Feedback	Learners' views on the scope, clarity, depth, and applicability of the feedback provided.
Human–AI Feedback Integration	The interaction and balance between AI-generated feedback and teacher feedback.
Pedagogical Misalignment	The mismatch between the system's feedback and the students' proficiency level, context, or instructional goals.
Design Improvement Suggestions	Suggestions from students and the instructor regarding the improvement of the system design.

Additionally, to ensure the accuracy and validity of the themes, the developed themes were presented to some students and faculty members for participant validation. This process supported the reliability and validity of the findings (Lincoln & Guba, 1985). During the coding and theme development process, the contextual integrity of the data was preserved to ensure content validity, and interpretations were structured in a manner consistent with the data.

Findings and Interpretation

This section presents data from semi-structured interviews with students and teachers through thematic analysis according to the research questions. The findings are structured within themes and sub-themes corresponding to each research question. Direct quotes from participants are included, and the analysis findings are interpreted.

Research Question 1: How does GAI-supported system affect the development of technical writing skills in EFL students; how do the strengths and limitations of these systems differ according to student and teacher experiences?

Overview of the process according to student level and teacher

Students were divided into three groups based on their participation in the assignments given to them during the research. Their perspectives and perceptions of the entire process, classified as High, Medium, and Low, were presented in Table 4, along with their frequencies, compared with those of other students and the teacher.

Table 4
Thematic code distribution according to student writing levels

	Coding System	High	Medium	Low	Teacher	Total
Development of Technical Writing Skill	Motivational Function of Feedback	5	3	5		13
	Revision-Oriented Writing					
	• Low Revision Engagement		1	2		3
	• Moderate Engagement with Feedback		2	1		3
	• Strategic Revision Behavior	3				3
	Draft Support	5	3		1	9
Perceptions of the Nature of Feedback	Immediate Feedback		1	1		2
	Lack of Explanation in Feedback	4	3	1	8	16
	Scope and Level of Feedback					
	• Lacking Macro Feedback	3			6	9
	• Micro Level Feedback	6	6	6	6	24
Human-AI Feedback Integration	Complementary Role of AI Feedback	3		1		4
	AI-Teacher Feedback Alignment	3	3	2		8
	Preservation of Teacher Feedback	2	4	1		7
Pedagogical Misalignment	Evaluation Concern				4	4
	Contextual Irrelevance					
	• ESL-EFL Mismatch				6	6
	Genre Expectation Gap				2	2
	Student Level - Appropriateness				6	6
Design Improvement Suggestions	Student Suggestions	8	7	2		17
	Teacher Suggestions				11	11
Total		42	33	22	50	147

The concept map presented in Figure 4 reveals the contributions and limitations of GAI-supported system technical writing skills in a multidimensional way based on the experiences of students and teachers.

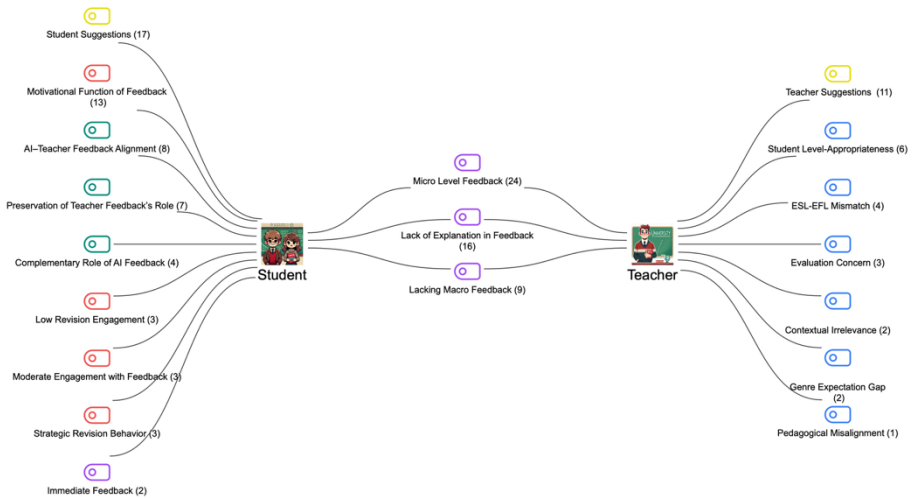


Figure 4. GAI-Supported system thematic map by participant type

The frequencies in the concept map show that the system effectively develops micro-level writing skills. However, it has macro-level support, clarity, contextual coherence, and pedagogical flexibility limitations. These findings suggest that when used with GAI-supported system teacher support, the system becomes more functional, thereby supporting the idea that a hybrid (AI + teacher) model may be more effective.

Research Question 2: What contributions does GAI-supported system make to the writing process of EFL students, and how do these systems provide the feedback perceived and utilised by students?

Theme 1: Development of technical writing skills

Student statements show that GAI-supported system contributes to technical writing skills, particularly in initial draft creation, identification of superficial language errors, and motivation support, but is limited in developing high-level revision skills.

Motivational function of feedback (n = 13)

Many students stated that the system's provision of immediate feedback encouraged them to engage in the writing process and increased their motivation to write:

"It showed me my mistakes. I tried to learn from them." (High-Level Student-3)

"Seeing that I made a lot of mistakes at first discouraged me a little. However, as I realised that I was improving over time, my motivation started to increase." (Medium-Level Student-1)

"The feedback was neither too detailed nor insufficient; however, in some cases, it bothered me quite a bit and made me nervous." (Low-Level Student-3)

Revision-oriented writing

Student views on revision revealed three different levels of engagement.

Low revision engagement (n = 3)

Some students stated that they only corrected errors indicated by the system and did not make any structural or content changes.

"I am not considering using it at the moment, but I may use it again if I find time during the summer holidays." (Low-Level Student-3)

Moderate engagement with feedback (n = 3)

Some students stated that they planned to use the system only when needed, for example, when there was no feedback from the teacher. The following student comment highlights the flexible use potential of the system, but also reflects a cautious approach to the extent to which the student will adopt and implement the feedback:

"I can use this system to improve myself and try out things I am curious about when I cannot get feedback from my teacher." (Medium-Level Student-1)

Strategic revision behaviour (n = 3)

A small number of students stated that they used feedback strategically and focused on making overall revisions to their text:

"It could be more useful for us because we will write a lot, for example, in the first year. And we need to see our mistakes. I think such an AI application could also benefit other classes." (High-Level Student-3)

Draft support (n = 9)

The system served as a guide for many students, especially during the initial draft stage of the writing process. Some students applied the feedback provided by the system step by step to improve their texts, reviewing their writing more carefully in the process. Students who used the system at a low level did not use it to support their drafts.

"After writing my paragraph, I was wondering where to start revising to improve my score. I would fix a conjunction, then a tense. But sometimes, if I corrected it incorrectly, my score would drop." (High-Level Student-1)

"Even in other courses, our speaking teacher asked us to write essays. I rewrote my essay draft in that way and revised it again." (Medium-Level Student-2)

Theme 2: Perceptions of the nature of feedback

Students' perceptions of the quality of feedback related to GAI-supported system were grouped under three main headings: Immediate Feedback, Scope and Level, and Lack of Explanation in Feedback. These headings reveal how students interpret feedback on the system and how they reflect it in the writing process.

Immediate Feedback (n = 2): Students stated that the system could be used to receive immediate Feedback, especially when they could not receive Feedback from the teacher or during individual work outside of class. However, only a minimal number of students emphasised this situation. Only n=2 coding was done to express this situation.

"Sometimes I did not have the opportunity to receive feedback in class because it was not my turn. In such cases, receiving immediate feedback from the system was useful." (Low-Level Student-2)

Scope and level of feedback

Lacking macro feedback (n = 9)

The system was limited in providing Feedback on higher-level writing skills such as paragraph organisation, coherence, and idea development. High-level students and teachers particularly criticised this.

'I expected more detailed explanations; for example, "You could use this word instead" or "The coherence of the topic is broken here". More comprehensive guidance on tenses and sentence structure would also be useful.' (High-Level Student-1)

"Grammar is essential in writing, but before that, there is critical thinking. Sentence variety and phrase combination are significant. Grammar mistakes can be fixed somehow. (Teacher)

Micro level feedback (n = 24)

The Feedback provided by the system is primarily focused on vocabulary, grammar, and spelling.

"At first, I was focused on punctuation marks in this system. That improved later. He also found my writing too short. Other than that, that was it." (Low-Level Student-1)

"...it shows grammar mistakes, but it often does not notice structural mistakes in phrases, for example." (Teacher)

Lack of explanation in feedback (n = 16)

Students did not understand the criteria used by the system to assign scores. Students at different levels of proficiency also supported this statement.

"... I could not understand how the score was calculated." (High-Level Student-1)

"Sometimes I could not fully understand what the system found wrong; I had difficulty figuring out what mistake I had made." (Medium-Level Student-1).

"No matter how carefully I wrote, I could not get the score I expected from the system. I did not understand why." (Low-Level Student-3)

"It gives 50 points for 'Language Accuracy,' but what does 50 mean? It is not clear what criteria this scoring is based on." (Teacher)

Research Question 3: What contributions does GAI-based system teacher feedback make to the writing instruction process when used?

Theme 3: Human–AI feedback integration

This theme examines how students compare teacher feedback with AI feedback, how feedback from these two sources integrates or complements or contradicts each other. It covers issues such as the extent to which AI feedback complements the teacher, how consistent it is, and which source students trust.

Complementary role of AI feedback (n = 4)

Some students indicated that the contribution of the AI-supported system to the writing process became more meaningful when combined with teacher feedback. These students stated that the system was effective not only as an independent tool but also as a complementary part of the teaching process.

"The software also contributed to the teacher's feedback." (High-Level Student-3)

"Both have an effect. Both the lessons and the software had an impact. I believe that." (Low-Level Student-1)

AI–Teacher feedback alignment (n = 8)

Some students stated that the feedback they received from the AI system was similar to teacher feedback and viewed this positively. This similarity increased students' desire to use the system more.

“At first, my teacher didn't like the essays I wrote, but after using the software, he started saying ‘it's good’ towards the end.” (High-Level student-2).

“No, there wasn't much of a difference. It was almost the same.” (Medium-Level student-2).

However, some students mentioned that there were differences between the two sources and that this sometimes caused confusion.

“Once, the software gave me a score of 40, but the teacher liked my paragraph.” (High-Level Student-3)

“... even though the teacher liked my paragraph, the software showed many more errors.” (Low-Level Student-2)

These findings show that students have both consistent and inconsistent feedback experiences.

Preservation of teacher feedback's role (n = 7)

Although students find the feedback they receive from GAI-supported systems valuable, they still consider teacher feedback indispensable. Especially in higher-level writing skills such as content development, coherence, and idea organisation, students evaluated teacher comments as more explanatory, guiding, and reliable.

“... The teacher's lessons really added a lot to me. I had never taken writing training before, and suddenly I had the opportunity to receive comprehensive training from a very good teacher.” (High-Level Student-1)

“...the face-to-face feedback from the teacher was more effective for me because I could understand my mistakes more clearly and directly.” (Medium- Level Student-1)

“.... what I learned from the teacher helped me improve a lot.” (Low-Level Student-2)

Research Question 4: What are the students' and teachers' suggestions for the design of a GAI-supported system?

In answering this research question, the shortcomings observed by the teacher were first categorised under the theme of 'Pedagogical Misalignment,' and suggestions for improving the system design were categorised under 'Design Improvement Suggestions>Teacher Suggestions.' In addition, students' views and suggestions

regarding the system design were included under 'Design Improvement Suggestions>Student Suggestions' to provide a multifaceted evaluation.

Theme 4: Pedagogical misalignment

This theme represents the teacher's design criticisms of the GAI-supported system and the points that they found pedagogically inadequate.

Evaluation concern (n = 3)

The teacher stated that the ambiguity of the system's scoring criteria weakened the reliability of the evaluation and that students tended to rely too much on digital tools in the online writing process.

"... If they write in Word and then paste it into Grammarly or Quillbot, they can correct the text with external editing tools." (Teacher)

Contextual irrelevance (n = 6)

The teacher criticised the system's inability to adapt to different contexts (e.g., EFL context). They stated that the system did not offer enough flexibility for the needs of students in Turkey.

"The material is suitable for ESL, but a version specific to the EFL context would be more useful and understandable for us." (Teacher)

Genre expectation gap (n = 2)

The teacher noted that the system does not focus enough on the structural expectations of genres such as essays and paragraphs, and therefore cannot guide student writing correctly.

"Students first learn sentence structures, then paragraph types. ... the system provides feedback directly on the essay type." (Teacher)

Student level-appropriateness (n = 6)

This code addresses the extent to which the feedback provided by the system is appropriate for the student's individual language proficiency, learning level, and stage in the academic writing process.

"This system seems to be designed for students who are doing advanced writing work such as academic writing or IELTS." (Teacher)

Theme 5: Design improvement suggestions

Teacher suggestions

1. "A similar version for EFL would be much better."

2. "It is necessary to limit the essay or paragraph, otherwise it is tough to read."
3. 'It would be good if students could see the assessment criteria on the screen.'
4. 'The criteria need to be clearly stated.'
5. 'If students can write and upload their work during class, that would be what I want.'
6. 'It is not clear what the errors are here. I would like to see them.'

Student suggestions

7. "More comprehensive feedback could be provided."
8. "I think the system could provide more detailed information... It would be good if it pointed out any issues with the coherence of the topic."
9. "...Just colour coding is fine, but changing the colour of the pen while writing the paragraph..."
10. "For example, when our mistakes appear, we could go back and underline the mistakes in the paragraph."
11. "... lists their suggestions below, but that part looks confusing. The words are mixed up. This part could be organised..."
12. "I think the scoring tool needs to be improved a little."

Discussion and Conclusions

This study aims to reveal the effects of GAI-supported system on English teacher education students' and writing teachers' perceptions of their writing skills, user experiences, and feedback processes. The findings indicate that system contribute significantly, particularly in technical components such as spelling, grammar, punctuation, and paragraph structure. The majority of students reported that they developed a more attentive and conscious attitude toward the writing process thanks to the immediate and score-based feedback they received from the system. These results are consistent with the role of systems in supporting technical writing skills, as Geng and Razali (2022) and Al-Inbari and Al-Wasy (2023) emphasized. These findings were examined under the theme of 'Development of Technical Writing Skills,' it was emphasised that systems are effective in teaching formal skills such as language accuracy and structural consistency.

However, the findings also revealed that the system is limited in supporting higher-level writing skills such as content production, idea development, and text organisation. Students noted that the system only identifies superficial errors (e.g., spelling or missing sentence warnings) but fails to explain their causes or provide guidance on correcting them. This finding can be addressed under the theme of 'Perceptions of the Nature of Feedback,' as the feedback provided by the system was described as lacking clarity, depth, and often inconsistent with student needs. This situation demonstrates that systems still require teacher support in writing processes involving cognitive complexity (Weigle, 2010; Yan, 2023). In particular, they cannot replace human evaluation in meaning construction, content consistency, and argument development (Barkaoui, 2011).

Another important finding obtained within the scope of the research is that, despite the technical functionality of the system, students sometimes find the feedback superficial and disconnected from the context. Students stated that the feedback provided by AWE system was general, while teacher feedback was more explanatory, directive, and pedagogically appropriate. This finding is consistent with the studies conducted by Guo and Wang (2024) and Su and Lai (2023). Furthermore, students' perception of feedback from teachers as more meaningful points to the need for the combined use of teacher and AI feedback under the theme of 'Human-AI Feedback Integration.' For students to have meaningful learning experiences, feedback should point out errors, explain why they were made, and how they can be corrected.

Data on students' use of the system outside of class also varies. While some students voluntarily use the AWE system to support the writing process, others interact with the system only in the context of mandatory assignments and assessments. This difference supports the finding of Tambunan et al. (2022) that individual motivation affects AWE system usage. This situation is evaluated under the theme of 'Student Level-Appropriateness,' as the system cannot provide differentiated feedback appropriate to each student's proficiency level. Students'

internalisation of the system as a learning tool will also facilitate their understanding of the feedback provided by the system.

On the other hand, teachers' views revealed that the system saves time, especially when reviewing initial drafts and conducting technical pre-assessments. Teachers noted that after the system addressed technical errors, they could focus more on content, structure, and idea development with students. In this context, AWE systems complement writing education with teacher feedback. However, teachers retain their central role in final assessment regarding pedagogical intuition, contextual analysis, and student-level guidance (Chiu et al., 2023). This is related to the theme of 'Preservation of Teacher Feedback's Role'; AWE systems cannot fully replace the teacher's role in guidance and decision-making in educationems.

Some important limitations of the AWE systems have also been identified. The system only allows text-based input and does not recognise handwriting, visual materials, or enriched text types. This makes it challenging to evaluate writing activities applied in face-to-face lessons. In addition, the system does not allow teachers to upload their assessment rubrics, which prevents the assessment process from being tailored to pedagogical goals. As evaluated under the 'Pedagogical Misalignment,' the system's predefined structure limits the teacher's ability to make flexible decisions appropriate to the context, level, and student needs. The system's rigid and predefined structure limits the teacher's ability to assess writing skills using different criteria, reducing pedagogical flexibility.

Furthermore, the system's structure, which is generally designed for an ESL (English as a Second Language) context, may lead to context-free or overly academic feedback for EFL students. This highlights the need for the system to be localised and level-sensitive. Based on all these findings, AWE systems have significant potential as a supportive tool in writing instruction; however, for this potential to be effectively utilised, the system must be structured to complement teacher guidance. System designs responsive to the needs of students and teachers, explanatory, flexible, and context-focused will contribute more comprehensively to developing writing skills.

Recommendations

Pedagogical Implications

- AWE systems can be used effectively in the final assessment stage of the writing process and cyclical stages such as drafting, reviewing, and rewriting.
- Students can be encouraged to use the system to obtain a score and improve their writing. In this context, students can be guided on how to read, interpret, and apply the feedback they receive from the system.
- The in-class use of the system can be diversified for different types of writing (e.g., descriptive, narrative, explanatory, etc.), thus supporting more flexible applications in genre-based writing instruction.

Recommendations for System Developers

- Customisable rubric infrastructures can be developed to enable teachers to define their assessment criteria and upload them to the system.
- This flexibility can make the system more suitable for the teaching context, writing type, and student level. Visual content recognition capabilities and compatibility with multiple data types (handwriting, screenshots, PDF, etc.) can be integrated into the systems. This allows the system to evaluate written production in person rather than limited to digital text.
- Feedback presentation can vary according to the student's level of development. More detailed explanations may be preferred for beginners, while shorter and more directive feedback may be preferred for advanced users.
- EFL context can localise systems. Feedback models can be developed to be sensitive to students' language levels, educational environments, and cultural differences.

Recommendations for Future Research

- This study was conducted with a limited sample and in a single context. Studies with different age groups, school types, and student groups with different writing levels may provide more generalisable results regarding the effectiveness of AWE systems.
- Mixed-method and long-term studies comparing teacher feedback with AWE systems feedback can be conducted. This will allow for a comparative assessment of the effects of different types of feedback on writing skills.
- The assessment performance of AWE systems can be examined in terms of different text types (e.g., narrative, descriptive, explanatory, argumentative), and the type-based sensitivity of the systems can be improved.

- In future studies, factors affecting student motivation, digital literacy level, and students' responses to feedback can be examined in greater depth using qualitative methods.

Araştırma ve Yayın Etiği

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Etik Kurul İzni

Atatürk Üniversitesi, Sosyal ve Beşeri Bilimler Etik Kurulu, 02.12.2022, Toplantı:13 Karar:32

Yazarların Katkı Oranı

Makaleye (1. yazar %60, 2. yazar %20 ve 3. Yazar %20) katkı sağlamıştır.

Çıkar Çatışması

Yazarlar arasında herhangi bir çıkar çatışması yoktur.

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Genişletilmiş Özet

Üretken Yapay Zeka (ÜYZ) dil eğitimine entegre edilmesi, öğrencilerin yazma becerilerini değerlendirme ve geliştirme konusunda dönüştürücü olanaklar sunmuştur. Bu uygulamalardan biri, öğrencilerin yazdıklarını anında değerlendirip geri bildirim sağlayan Otomatik Metin Değerlendirme (OMD) sistemidir. Bu sistemler EFL (Yabancı Dil Olarak İngilizce) bağlamında giderek daha fazla kullanılmasına rağmen, pedagojik uyumu, geri bildirim kalitesi ve bağlamsal uygunluğu konusunda sorular hala devam etmektedir. EFL öğrencileri için yazma becerilerinin değerlendirilmesi, yalnızca dilsel doğruluk değil, aynı zamanda eleştirel düşünme, yapı ve içerik geliştirme açısından da bütünsel bir yaklaşım gerektirir. Bu çalışma, ÜYZ destekli bir OMD sisteminin EFL öğretmen adayları ve eğitimcilerinin yazma gelişimleri, geri bildirim deneyimleri ve öğretim algıları üzerindeki etkisini araştırmaktadır.

Bu nitelik durum çalışması; İngilizce Öğretimi (ELT) programından 53 lisans öğrencisinden seçilen 9 öğrenci ve bir yazma eğitimcisi kapsamaktadır. OMD sistemi, OpenAI'nin API'sini kullanarak geliştirilmiş Virtual Writing Tutor uygulamasına ait Moodle eklentisidir. Öğrenciler, çok sayıda yazma görevi aracılığıyla sistemle etkileşime girmiş ve gramer, noktalama ve yapısal konular hakkında otomatik geri bildirim almıştır. Katılımcılar, etkileşim sıklıklarına göre Yüksek, Orta ve Düşük katılım gruplarına ayrılmıştır. Bu katılımcı seçimi, sistemin farklı etkileşim düzeylerinde nasıl algılandığını karşılaştırmalı olarak değerlendirebilmek için yapılmıştır. Veriler, yarı yapılandırılmış görüşmeler yoluyla toplanmış ve Braun ve Clarke'ın (2006) tematik analiz yaklaşımı kullanılarak analiz edilmiştir.

Teknik Yazma Becerilerinin Gelişimi; OMD sistemi, öğrencilerin yazma becerilerinin teknik yönlerini geliştirmede özellikle etkili olmuştur. Katılımcıların çoğu gramer, imla, noktalama ve paragraf yapısında iyileşme olduğunu bildirmiştir. Anında geri bildirim, öğrencilerin mekanik doğruluğa daha fazla dikkat etmelerini ve çalışmalarını gözden geçirmeleri için onları motive etmiştir. Bu bulgu, yapay zeka araçlarının yüzeysel hataları tespit ederek yazmanın biçimsel yönlerini destekleyebileceği yönündeki önceki bulgularla uyumludur (Geng & Razali, 2022). Öğrenciler kendilerini daha bağımsız ve temel hataları düzeltebilecek durumda hissettiler, bu da teknik yazma becerilerinin gelişmesine katkıda bulundu.

Geri Bildirimin Doğası Hakkında Algılar; OMD sistemi dil doğruluğuna yardımcı olsa da geri bildirimi genellikle belirsiz, genel ve açıklayıcı derinliği eksik olarak algılandı. Öğrenciler, sistemin sorunları tespit ettiğini ancak özellikle tutarlılık veya argümantasyon gibi üst düzey yazma sorunları için uygulanabilir çözümler sunmadığını belirtti. Bu durum, sistemin biçimlendirici bir öğrenme aracı olarak algılanan yararını azalttı. Katılımcılar, hataları gösteren ve net düzeltme önerilerinde bulunan geri bildirimler istedi.

İnsan-YZ (Yapay Zeka) Geri Bildirim Entegrasyonu; Öğrenciler ve eğitmen, YZ tarafından oluşturulan geri bildirimlerin tek başına etkili öğrenme için yetersiz olduğunu vurguladı. Sistem anında yanıt verse de insan geri bildirimlerinin derinliği, detayları ve pedagojik uyumu ile eşleşemiyordu. Öğretim elemanı; ilk taslaklar için, özellikle teknik sorunları belirlemede yararlı buldu. Ancak içerik, fikir geliştirme ve retorik yapı için kendi geri bildirimlerine güvendi. OMD sisteminin düşük düzeydeki düzeltmeleri, öğretmenin ise karmaşık, bağlamsal ve kavramsal yönleri ele aldığı bir hibrit model, teknolojinin en etkili kullanımı olarak görüldü.

Pedagojik Uyumsuzluk; Güçlü yönlerine rağmen, OMD sistemi gerçek öğretim ve öğrenim bağlamıyla uyum konusunda birkaç sınırlama gösterdi. İlk olarak, sistem katıydı ve öğretmene özelleştirilmiş değerlendirme ölçekleri yükleme veya puanlama kriterlerini ayarlama seçeneği sunmuyordu. İkinci olarak, ESL odaklı tasarımı, akademik üslup, kültürel uygunluk ve öğrenci seviyesi açısından EFL öğrencileri için uygunsuz hissedilen geri bildirimlere yol açtı. Bu sorunlar, sistemin yerelleştirme ve esneklik eksikliğini ortaya çıkardı ve dersin öğretim hedefleri ve öğrenci ihtiyaçları ile pedagojik uyumluluğunu azalttı. Bu sınırlılıklar, özellikle öğretim programına entegre edilmiş yazma kazanımlarının sistem tarafından tam olarak desteklenememesi sorununu beraberinde getirmiştir.

Tasarım İyileştirme Önerileri; Katılımcılar, OMD sisteminin pedagojik yararını iyileştirmek için belirli önerilerde bulundu. Bunlar arasında şunlar yer aldı:

- Bir hatanın neden sorunlu olduğunu ve nasıl düzeltilebileceğini açıklayan bağlamsal geri bildirim,
- Öğretmenlerin sistem çıktısını kurs hedefleriyle uyumlu hale getirmelerine olanak tanıyan rubrik özelleştirme seçenekleri,
- Genel ESL modellerine dayanmak yerine, geri bildirim EFL öğrencilerinin ihtiyaçlarına göre uyarlayabilmek için yerelleştirme özellikleri,
- Fikir geliştirme ve retorik yapı için biçimlendirici yorumlar dahil olmak üzere esnek geri bildirim türleri,

Bu öneriler, öğretim kararlarını destekleyen ve öğrenci çeşitliliğini barındıran, daha duyarlı, öğretim elamanın kolay kullanabileceği OMD ihtiyacını pekiştirmektedir.

ÜYZ destekli OMD sistemleri, teknik doğruluğu artırarak, öğretmenlerin iş yükünü azaltarak ve öğrencilerin kendi kendilerini düzeltmelerini teşvik ederek EFL yazma öğretimine önemli ölçüde fayda sağlayabilir. Ancak, geri bildirim derinliği, uyarlanabilirlik ve pedagojik uyumdaki mevcut sınırlamalar, bu tür sistemlerin öğretmen geri bildirimine yerine geçmekten ziyade tamamlayıcı araçlar olarak kullanılması gerektiğini göstermektedir.

Eğitim potansiyellerini tam olarak gerçekleştirebilmek için OMD sistemleri:

- Biçimlendirici, şeffaf ve öğrenciye duyarlı geri bildirim sağlamalıdır,
- Öğretmenlerin özelleştirme ve pedagojik kontrolüne izin vermelidir,
- Yerel öğrenme bağlamları ve EFL gerçekleri göz önünde bulundurularak tasarlanmalıdır.

Araştırma, ÜYZ destekli yazma değerlendirme araçlarının sadece teknik değil, aynı zamanda pedagojik düzeyde nasıl geliştirilebileceğine dair öneriler sunmaktadır. Gelecekteki araştırmalar, öğrenci profillerine dinamik olarak uyum sağlayan ve öğretim uygulamalarıyla daha derin bir şekilde entegre olan uyarlanabilir OMD sistemlerinin geliştirilmesine odaklanabilir. Öğretmenler, öğrenciler ve geliştiricilerin katılımıyla gerçekleştirilen işbirliğine dayalı tasarım çabaları, bu araçların teknolojik olarak gelişmiş ve pedagojik olarak etkili olmasını sağlayabilir. Sonuç olarak, OMD araçlarının EFL eğitiminde gelecekteki başarısı, teknik yeteneklerine ve çeşitli öğrenenlerin pedagojik hedefleri ve bağlamsal gerçeklikleriyle ne kadar uyumlu olduklarına bağlıdır.